



**FMEDA Development
White Paper
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FMEDA – Accurate Product Failure Metrics

Introduction

The letters FMEDA form an acronym for “Failure Modes Effects and Diagnostic Analysis.” The name was given by Dr. William M. Goble in 1994 to describe a systematic analysis technique that had been in development since 1988 by engineers now at *exida* to obtain realistic, accurate reliability metrics that match field failure results. These metrics include device / product level failure rates, failure modes, diagnostic capability, proof test effectiveness, and useful life. (Figure 1).

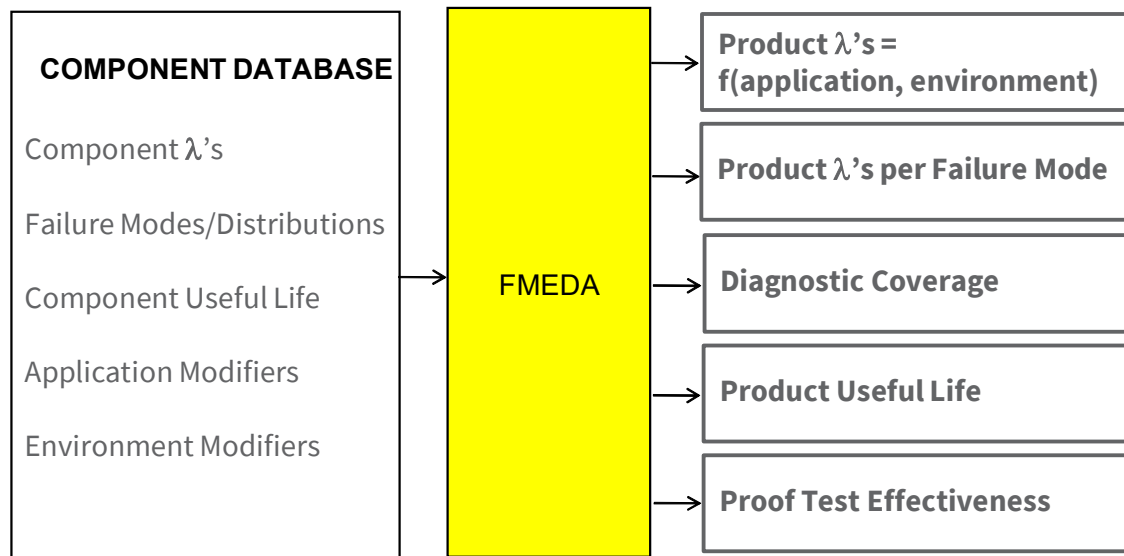


Figure 1: FMEDA Data Inputs and Data Outputs.

FMEDA Development

The FMEDA method was developed based on a combination of techniques. The initial FMEDA used a foundation of the FMEA analysis process based on a paper in the 1984 RAMS Symposium [COL84]. This expanded FMEA was further enhanced to add quantitative information needed to perform functional safety performance analysis. The first two pieces of information added were the quantitative failure rate and the distribution of failure modes for all components being analyzed. The third piece of information added to an FMEDA is the probability of the system or subsystem to detect the specific identified component failure mode via automatic on-line diagnostics or proof test.



The need to quantitatively measure diagnostic capability was recognized in the late 1980's [AME87]. In that context the basic concept for the modern FMEDA were first documented in the book *Evaluating Control System Reliability* [GOB92]. The method was enhanced further and the term FMEDA was first used in 1994 [MOR94]. After further refinement, the methods were published in the late 1990's [GOB98a, GOB98b, GOB99]. The method was explained to members of the IEC 61508 committee in the late 90s and included in Part 6 of the standard as a method of determining failure rate, failure mode, and diagnostic coverage for products.

The FMEA Foundation

A Failure Modes and Effects Analysis, FMEA, is a structured qualitative analysis of a system, subsystem, process, design, or function to identify potential failure modes, their causes, and their effects on (system) operation.

The concept and practice of performing a FMEA, has been around in some form since the 1960's. The practice was first formalized in 1970s with the development of US MIL STD 1629/1629A.

The FMEA method has proven to be cost effective, because design issues are detected early in development process saving costly rework. In early practice its use was limited to select applications and industries where cost of failure was particularly high. The primary benefits were to qualitatively evaluate the safety and reliability of a system, determine unacceptable failure modes, identify potential design improvements, plan maintenance activities, and help understand system operation in the presence of potential faults. Today the method is fully supported by engineering tools (Figure 2).

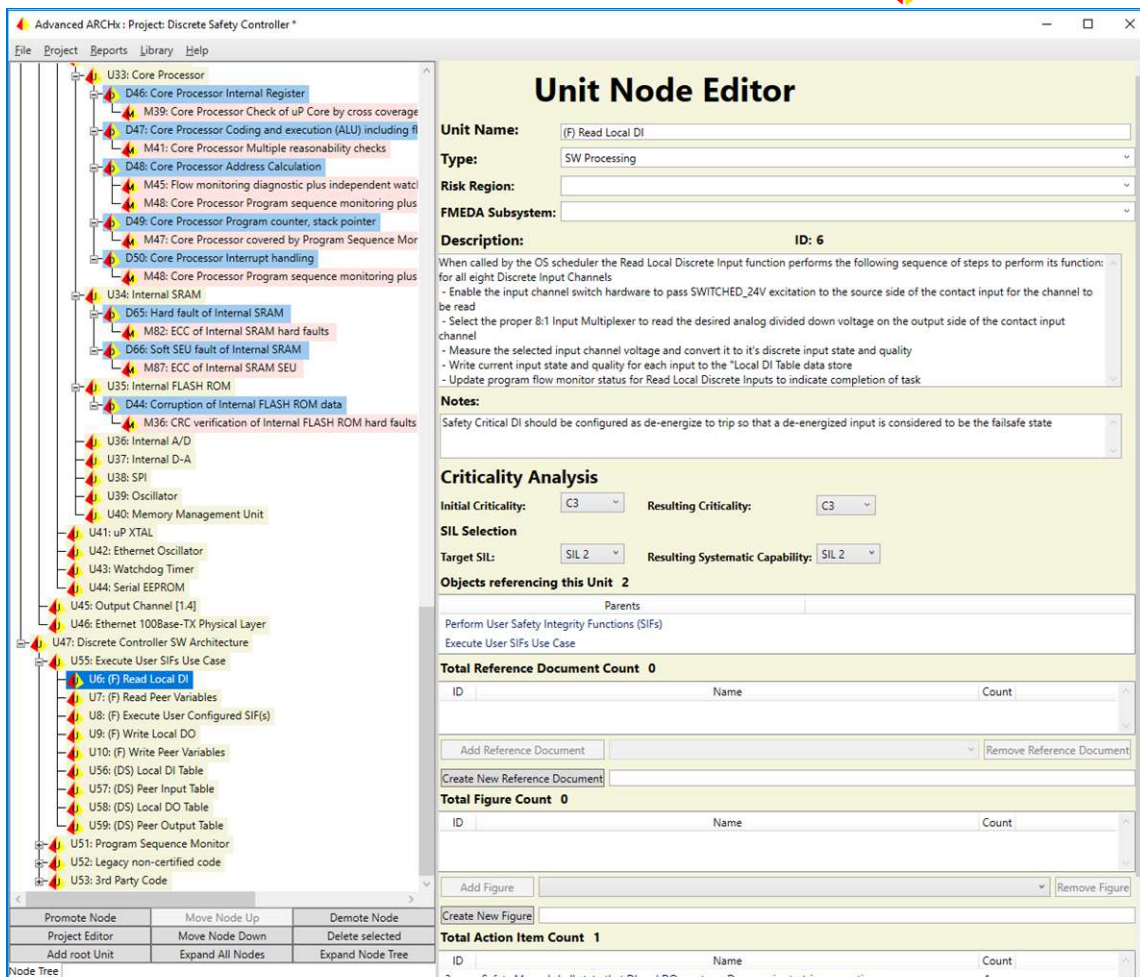


Figure 2: ARCHx™ FMEA Tool from *exida* Innovation LLC.

The FMEDA Technique

The FMEDA technique considers

- All components of a design,
- The functionality of each component,
- The failure modes of each component,
- The Useful Life of each component,
- The impact of each component failure mode on the product functionality,
- The design strength (de-rating, safety factors),
- The operational profile (environmental stress factors),
- The ability of any automatic diagnostics to detect the failure, and
- The ability of any manual proof test to detect the failure.

For an electronic FMEDA, the engineering inputs are a schematic, bill of material, and component specifications for any special parts. For a mechanical FMEDA, the engineering inputs are the assembly drawings, bill of material, component specifications, and any other drawings needed to understand how the device works. For all FMEDAs, the automatic diagnostic descriptions are needed. And for those FMEDAs where proof testing can be done, the proof test procedure(s) are required (Figure 3).

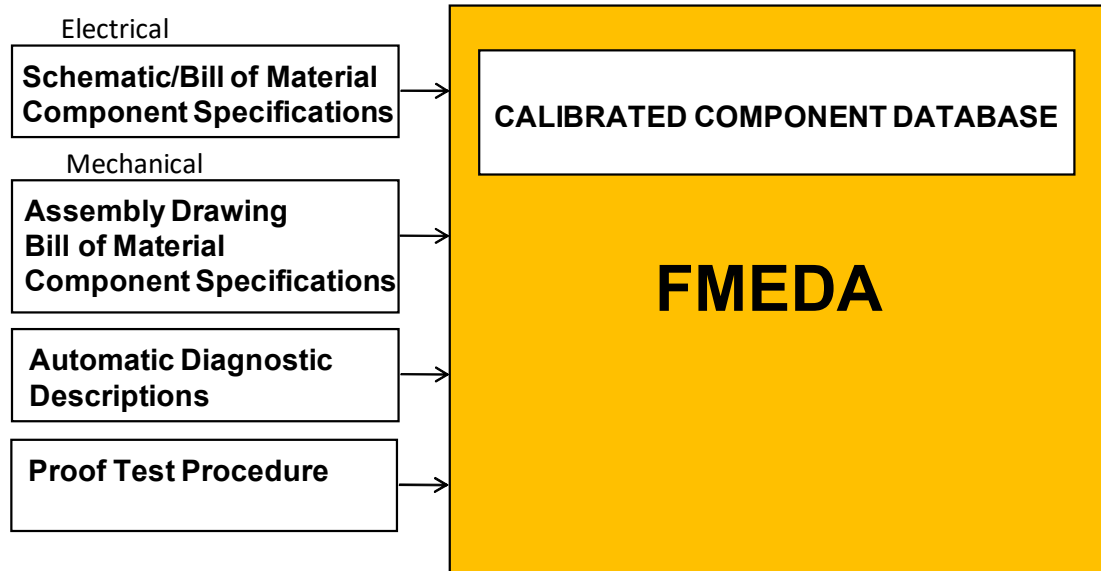


Figure 3: FMEDA Engineering Inputs.

In the FMEDA process, each component and its failure modes are listed. The ability of the automatic diagnostics to detect that particular component failure mode is estimated. And the ability of any specified proof test to detect the particular component failure mode is estimated. Normally an FMEDA tool references a component database where the component useful life and failure rates for each failure mode are accessed. The calculations can be completely done when all components are entered. Some tools, like *exida's* FMEDAx, calculate after each component is entered (Figure 4).

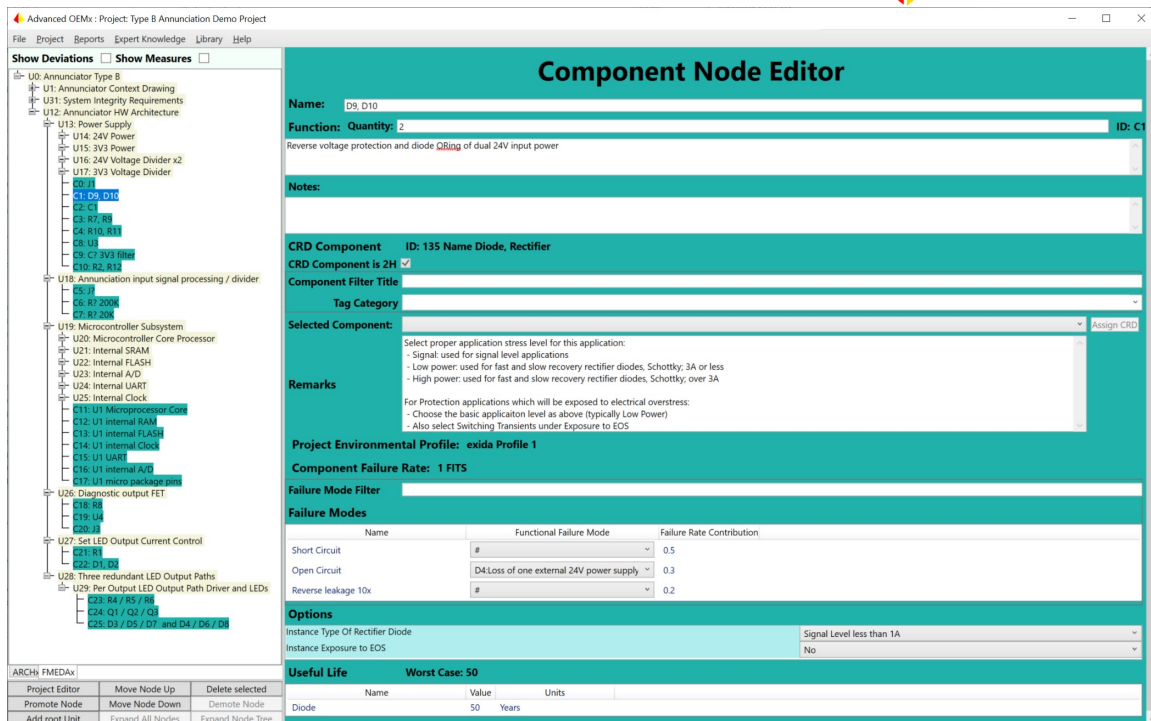


Figure 4: Component Node Editor screen for FMEDAx™.

FMEDA Evolution

FMEDA techniques have been further refined during the 2000's primarily during IEC 61508 preparation work. The key changes have been:

1. Use of Functional Failure Modes;
2. Mechanical Component Usage;
3. Prediction of manual proof test effectiveness; and
4. Prediction of product useful life.

With these changes, the FMEDA technique has matured to become more comprehensive and useful.

Functional Failure Mode Analysis

In the early 2000's functional failure mode analysis was added to the FMEDA process by John C. Grebe. In early FMEDA work, component failure modes were mapped directly to "safe" or "dangerous" categories per IEC 61508. This was relatively easy since everything that was not "dangerous" was "safe." It was recognized that these definitions did not predict accurate device safe failure rate as some failures caused loss of diagnostics but not safe failure [GOB01]. *exida*



added the “Annunciation” failure mode. With multiple failure mode categories now existing, direct assignment became increasingly difficult. In addition, it became clear that the category assignment might change if a product were used in different applications. With direct failure mode category assignment during the FMEDA, a new FMEDA was required for each new application or each variation in usage.

Using the functional failure mode approach, the actual functional failure modes of each subsystem are identified during an FMEA. During the FMEDA, each component failure mode is mapped to a product functional failure mode. The functional failure modes of the product are then categorized according to safe, dangerous, annunciation, etc. in a particular application. For any new application, only the final step needs to be repeated. This approach also facilitates reuse of subsystem analysis when existing circuits are used in new products.

Mechanical FMEDA Techniques

It became clear in the early 2000’s that many products being used in safety critical applications had mechanical components. An FMEDA done without considering these mechanical components was incomplete, misleading, and potentially dangerous. The fundamental problem in using the FMEDA technique was the lack of a mechanical component database that included part failure rates and failure mode distributions.

Using a number of published reference sources, *exida* began development of a mechanical component database in 2003 [GOB03]. After a few years of research and refinement [GOB07], the database has been published [EXI06]. This has allowed the FMEDA to be used on combination electrical / mechanical components and purely mechanical components.

Manual Proof Test Effectiveness

The FMEDA can predict the effectiveness of any defined manual proof test in the same way it can predict automatic diagnostic coverage. An additional column is added to the FMEDA and the probability of detection for each component failure mode is estimated. The cumulative effectiveness of the proof test is calculated in the same way as automatic diagnostic coverage.

Product Useful Life

Product useful life is the time period when the failure rate is reasonably constant. The useful life ends when the failure rate starts to significantly increase. For a given device, the useful life is limited by the component(s) with the shortest useful life period. If the component database contains useful life limitations for

each component as a function of operations conditions, then the useful life of a product can be predicted.

The Component Database

FMEDA results depend on the component database. Since the objective of the entire method is to obtain realistic, accurate reliability metrics that match field failure results (Figure 5), the component database is the most important part of any FMEDA.

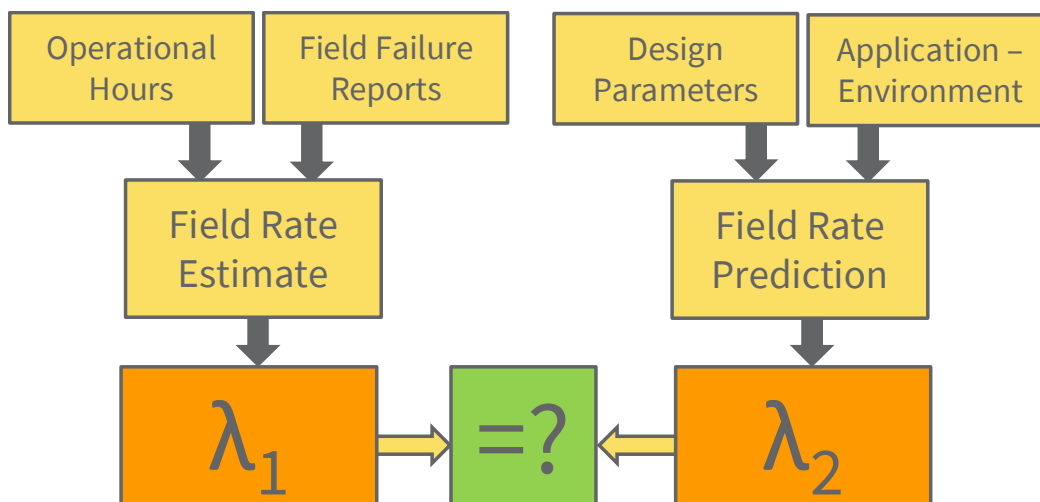


Figure 5: FMEDA Predictions should match Field Failure Estimations.

To ensure this level of accuracy, the *exida* Component Reliability Database (CRD)[™] has validated with hundreds of field failure comparisons. Using several sources including:

- 1) Field failure data from manufacturers,
- 2) the SILStat[™] operational event capture tool data,
- 3) published end user failure studies, and
- 4) industry databases such as the OREDA database [ORE15]

FMEDA predictions are compared to the field failure results in a “closed loop” manner. Any discrepancies are studied and root cause analysis of failures is used to adjust the component parameters and formulas to resolve the discrepancy. New component failure modes are discovered. New application stress factors have been discovered. When results match, the component database is validated [BUK07, BUK09, BUK17, EXI21b].

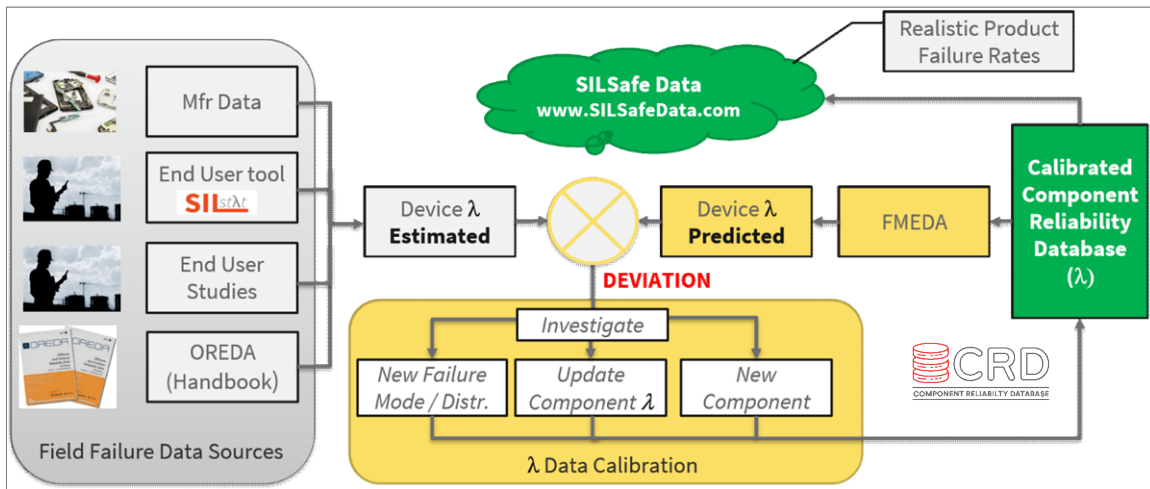


Figure 6: Closed Loop Field Failure Validation of Component Database.

The comparisons and validations cannot stop. Component designs of many types are improving and achieve lower failure rates. Given the constant development of better technology, continuous refinement of the component database with selective calibration to different operation profiles is required for accuracy. New variables that impact the reliability metrics will be identified. The component database must be regularly updated.

Conclusion

Given a component database calibrated with field failure data that is reasonably accurate [EXI21a], the FMEDA method has evolved into an effective tool that can predict realistic product level failure rate and failure mode data for a given application and environment. The predictions have been shown to be more accurate [GOB14] than field warranty return analysis or even typical field failure analysis given that these methods depend on reports that typically do not have sufficient detail information in failure records [GOB09, EXI21b].

As more data becomes available, the component database can be refined and updated as required by new technology and new knowledge. The success of the FMEDA technique in supplying needed data in a relatively accurate way has allowed the probabilistic, performance approach to design to work.



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Revision History

Rev. 1.0,	Goble, Grebe	Initial release, February 19, 2007
Rev. 1.1,	Goble, Grebe	Review results included, February 19, 2007
Rev. 1.2,	Goble, Grebe	Update, October 22, 2009
Rev. 2.0,	Goble, Grebe	Update, October 11, 2021



exida – Who we are.

exida is one of the world's leading accredited certification and knowledge companies specializing in automation system cybersecurity, safety, and availability. Founded in 2000 by several of the world's top reliability and safety experts, exida is a global company with offices around the world. exida offers training, coaching, project-oriented consulting services, standalone and internet-based safety and cybersecurity engineering tools, detailed product assurance and certification analysis, and a collection of online safety, reliability, and cybersecurity resources. exida maintains a comprehensive failure rate and failure mode database on electrical and mechanical components, as well as automation equipment based on hundreds of field failure data sets representing over 350 billion unit operating hours.

exida Certification is an ANSI (American National Standards Institute) accredited independent certification organization that performs functional safety (IEC 61508 family of standards) and cybersecurity (IEC 62443 family of standards) certification assessments.

exida Engineering provides the users of automation systems with the knowledge to cost-effectively implement automation system cybersecurity, safety, and high availability solutions. The exida team will solve complex issues in the fields of functional safety, cybersecurity, and alarm management, like unique voting arrangement analysis, quantitative consequence analysis, or rare event likelihood analysis, and stands ready to assist when needed.

Training

exida believes that safety, high availability, and cybersecurity are achieved when more people understand the topics. Therefore, exida has developed a successful training suite of online, on-demand, and web-based instructor-led courses and on-site training provided either as part of a project or by standard courses. The course content and subjects range from introductory to advanced. The exida website lists the continuous range of courses offered around the world.

Knowledge Products

exida Innovation has made the process of designing, installing, and maintaining a safety and high availability automation system easier, as well as providing a practical methodology for managing cybersecurity across the entire lifecycle. Years of experience in the industry have allowed a crystallization of the combined knowledge that is converted into useful tools and documents, called knowledge products. Knowledge products include procedures for implementing cybersecurity, the Safety Lifecycle tasks, software tools, and templates for all phases of design.

Tools and Products for End User Support

- exSILentia® – Integrated Safety Lifecycle Tool
 - PHAx™ (Process Hazard Analysis)
 - LOPAx™ (Layer of Protection Analysis)
 - SILAlarm™ (Alarm Management and Rationalization)



- SILect™ (SIL Selection and Layer of Protection Analysis)
- Process SRS (PHA based Safety Requirements Specification)
- SILver™ (SIL verification)
- Design SRS (Conceptual Design based Safety Requirements Specification)
- Cost (Lifecycle Cost Estimator and Cost Benefit Analysis)
- PTG (Proof Test Generator)
- SILstat™ (Life Event Recording and Monitoring)
- exSILentia® Cyber- Integrated Cybersecurity Lifecycle Tool
 - CyberPHAx™ (Cybersecurity Vulnerability and Risk Assessment)
 - CyberSL™ (Cyber Security Level Verification)

Tools and Products for Manufacturer Support

- FMEDAx (FMEDA tool including the exida EMCRH database)
- ARCHx (System Analysis tool; Hardware and Software Failure, Dependent Failure, and Cyber Threat Analysis)

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